

Project Econometric - Did the COVID-19 pandemic change the income elasticity of health expenditure in OECD countries?

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1 Introduction

Healthcare expenditure has risen significantly over recent decades, driven by the aging population, technological advances in medical care, and the enlargement of the public coverage system. According to OECD (2025), OECD countries spent on average nearly 9.3% of GDP on health in 2024. Although this is lower than the peak of the COVID-19 period, it remains higher than the pre-pandemic level of 8.8% of GDP recorded in 2019 (OECD, 2023). Moreover, 16 OECD member states currently exceed a 10% threshold for health expenditure as a share of GDP. Among the highest-spending nations, the United States reaches 17.2%, which is approximately double the OECD average. Therefore, evaluating the income elasticity of health expenditure—specifically whether it is classified as a luxury good or a necessity good within a state scheme—is essential for shaping effective public policy.

Newhouse (1977) has already stated that health expenditure behaves as a luxury good, which means it rises more than proportionally with the income rise in his cross-section OECD data. As a consequence, income is considered the main factor to accelerate cross-country differences. On the other hand, subsequent panel data research for OECD countries advocates the idea that health expenditure is a necessity good associated with income elasticity, which is closer to one or slightly above one. Sen (2005) examined the connection between income per capita and health expenditure using data in the 1990–1998 period for 15 OECD countries. While controlling for additional cost and supply variables, the paper indicated that income elasticity varies in the range from 0.21 to 0.51. Later studies by Dreger and Reimers (2005) using a panel co-integration techniques approach and data from 21 OECD countries

showed that health care expenditure is not a luxury good. Chakroun (2010) conducted research with a sample of 17 OECD countries in the period 1975–2003 with a threshold regression model; the income elasticity is below unity for the tested countries, with a range from 0.6 to 0.87. However, the income elasticity varies between countries and institutional operations, suggesting that health systems may impact the relationship between income and health expenditure. It highlights the condition to assess the income elasticity through the cross-country analysis instead of focusing only on a small group of countries.

The coronavirus disease (COVID-19), caused by the SARS-CoV-2 virus, was first reported in Wuhan, China, in December 2019 and quickly spread worldwide, becoming a global crisis. It provided a simultaneous shock to the public health system and national economies across OECD countries. Governments intensely dedicated resources to health spending to develop medical care capacity, field hospitals, and vaccine supply chains while GDP decreased steeply. Countries that significantly expanded their public health systems during the pandemic may now face pressure to maintain health spending even if the growth rate does not increase correspondingly. This suggests that the income elasticity of health expenditure post-pandemic could fluctuate, complicating the estimation of the relationship between health spending and GDP. The study examines the impact of COVID-19 on the income elasticity of health expenditure across 38 OECD countries. The research tests the responsiveness of health spending to GDP per capita after the pandemic, using comprehensive panel data for the 2000–2024 period. The analysis covers both cross-country and within-country variation via fixed effects with a COVID interaction term. While a few papers inspect structural breaks around COVID-19 using panel data, recent OECD reports primarily display spending growth descriptively, without exploiting econometric models. This paper contributes the first analysis comparing the change in income elasticity between the pre- and post-COVID-19 periods across OECD countries.

The remainder of this paper is structured as follows: Section II reviews the associated literature on the income elasticity of health expenditure. Section III describes the dataset, variable definitions, and sample selection criteria. Section IV provides a descriptive analysis of the main variables and evidence of the potential structural break around 2020. Section V presents the econometric model, discusses the choice of controls, and addresses endogeneity concerns. Section VI discusses the main empirical results. Section VII reviews the robustness analysis. Section VIII concludes with a discussion of the findings, limitations, and policy implications.

2 Literature Review

The study assessing the relationship between income elasticity and health expenditure was pioneered by the seminal work of Newhouse (1977). By estimating a cross-sectional model of 13 OECD countries, he found that income explained over 90% of cross-country variation in medical expenditure. His results demonstrated that the elasticity exceeds unity, meaning health expenditure behaves as a luxury good. As a result, high-income countries allocate a larger share of GDP to medical matters, with this spending rising faster than income. This conclusion is aligned with Gerdtham (1992), who examined the regression of healthcare spending on GDP while controlling for institutional and socio-demographic factors. Applying data from 20 OECD countries from 1960–1987, they also found that healthcare is a luxury good. However, using the same dataset, Hitiris and Posnett (1992) tested the model while accounting for unobserved heterogeneity through country-specific effects. Their results indicated an income elasticity close to one, contradicting the view that healthcare is a luxury good. Dreger and Reimers (2005) reached a similar conclusion using panel cointegration techniques on 21 OECD countries during 1975–2001, further indicating that health expenditure is not a luxury good. Later studies have focused on the introduction of heterogeneity and nonlinearity. Chakroun (2010) utilized a panel threshold regression model for 17 OECD countries over the period 1975–2003 and found that healthcare behaves as a necessity good on average, characterized by a dynamic nonlinear relationship. The argument for health as a near-necessity is also highlighted in the research of Murthy and Okunade (2009). They inspected cross-sectional data with heterogeneous cointegration and found that long-run elasticities varied between 0.85 and 1. These panel tests consistently suggest that income elasticity is below or close to unity in the long run when acknowledging the existence of fixed effects and covariates such as the aging population and technology diffusion. Getzen (2006) highlights that the level of aggregation and the time dimension significantly impact income elasticity. At the individual household level, the income elasticity is typically very low, whereas at the national budget level, it tends to be higher. Casas et al. (2021) further emphasized the importance of time variation, noting that while elasticity estimates are often higher in cross-sectional analyses, they are typically bounded within the interval 0.7–0.9 in dynamic panels. This reflects the fact that public health financing systems often respond slowly and do not immediately catch up with changes in the income growth rate.

The COVID-19 pandemic triggered an unexpected disruption to the link between income and health expenditure across OECD countries. According to OECD (2021), health expenditure grew by approximately 5% in 2020, while average real

GDP across the OECD fell sharply by 4.5%, creating the largest historical gap between income and health spending. Authorities dedicated resources to rapidly expand hospital capacity, testing, and vaccine procurement based on public health demand, largely without the typical restrictions of national budgets or fiscal constraints. Khan et al. (2016) analysed healthcare expenditure and economic growth causality in South Asian Association for Regional Cooperation (SAARC) countries using panel cointegration and causality methods for the 1995–2012 period, employing GDP, the labour force, literacy rates, and the elderly population (65+) as economic indicators. Their findings show that the income elasticity of healthcare expenditure in South Asian countries is less than unity in both the short- and long-run, indicating that healthcare spending increases more slowly than income growth.

At present, there is no prior study examining how the COVID-19 pandemic impacts income elasticity using panel data extending into the post-pandemic period across all OECD member states. The existing empirical literature relies predominantly on data ending before 2020, while studies of the COVID-19 fiscal response focus on absolute spending levels rather than the income–spending relationship. This paper fills that gap directly. Using panel data for OECD countries over the 2000–2024 period, this study provides insight into the relationship between health expenditure and GDP per capita. The model examines how expenditure responded to changes in income and whether healthcare behaved as a luxury or a necessity good.

3 Dataset and sample selection

3.1 Variables and data sources

All variables are extracted from the primary source: the OECD Health Statistics database, accessed through the OECD Data Explorer.

Health expenditure is the dependent variable, measured in US dollars at constant prices and purchasing power parity (PPP) as total current health expenditure per capita. The PPP conversion ensures that price level differences across countries do not distort the comparison of actual spending. GDP per capita is the primary explanatory variable, representing the level of economic activity and income for each country per year. This variable is measured in constant prices (chain-linked volumes) and specified in US dollars at 2015 PPP. Chain-linking prevents price shifts from changing the base year, while the PPP conversion maintains a real income level that is comparable across nations. Both variables are transformed into natural logarithms ($\ln$) to allow for the interpretation of coefficients as elasticities and to minimize the impact of skewness in the data. Beyond these two core variables, the dataset

introduces a control variable to account for other determinants of health spending. The demographic variable is defined as the percentage of the population aged 65 and over. Given that older populations require more intensive healthcare and medical treatment, separating this variable is essential to correctly estimate the isolated effect of aging on healthcare expenditure.

3.2 Sample selection and country coverage

The study selects OECD countries solely due to three main reasons. First, member states have a common institution, and the data quality and measuring method of health spending could lead to the comparison through these countries being more reliable. Second, restricting samples to OECD countries brings similar-income economies with developed health systems to ensure income elasticity could be meaningful and comparable. Third, mixing low and middle-income countries in different health systems, institutional capacity, and COVID-19 response would make the estimation result incorrect. The sample period begins from 2000 to capture sufficiently pre-pandemic dynamics associated with the relationship between health spending and GDP, and lasts for 24 years to provide the post-pandemic recovery period. The time includes the COVID-19 years to examine how changes in income elasticity occurred before and after the pandemic. Because of incomplete data for countries for every year, the missing values are excluded to prevent artificial trends.

4 Descriptive Analysis

4.1 Summary statistics

Table 1 demonstrates summary statistics for the main variables across 38 OECD countries with 913 observations between 2000 and 2024. The health expenditure is \$3,842 on average, within the range from \$592 in early 2000 to \$11,926 in recent years. GDP per capita fluctuates among the countries, ranging from \$10,904 to \$130,164, with a mean of \$46,217, suggesting that incomes could be impacted by heterogeneous effects and reflecting the significant income gap between high-income OECD countries and lower-income countries. There are 15.76% people who are aged 65 and over, varying from young population countries such as Mexico and Colombia to the ageing countries such as Japan and Italy, with the percentage 5.35% and 29.27% respectively. These figures state that the within and between-country variation included is sufficient to estimate the model using the approach of fixed-effects described in full detail in section V.

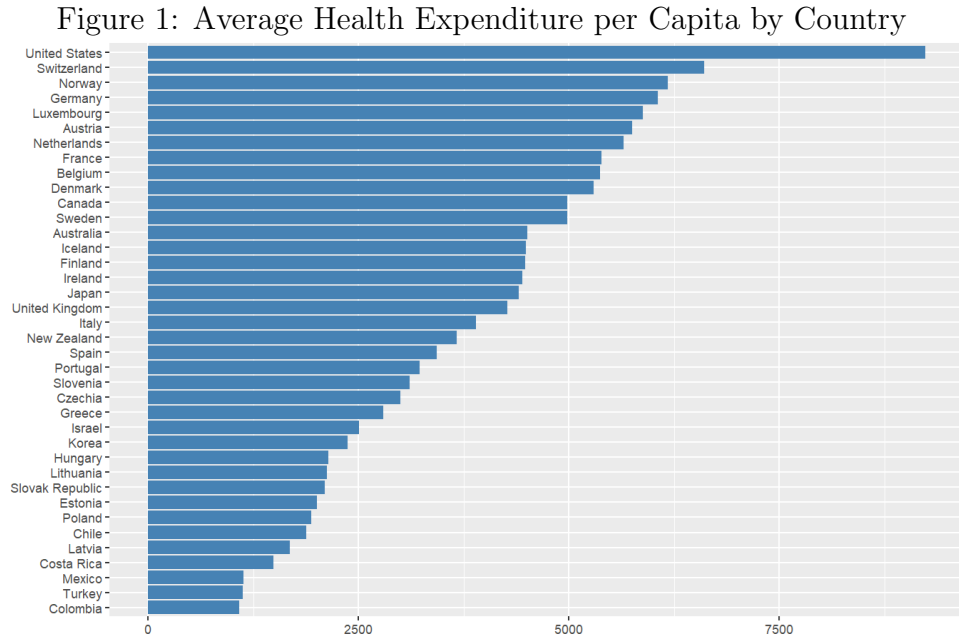
Table 1: Summary Statistics of 38 OECD Countries in period 2000–2024)

Variable	Obs.	Mean	Median	Min	Max
Health Expenditure per Capita (USD PPP)	913	3,842.5	3,756.2	592.2	11,926.4
GDP per Capita (USD PPP)	913	46,217	45,725	10,904	130,164
Population Aged 65+ (%)	913	15.76	16.20	5.35	29.27

Source: Own compilation.

4.2 Cross-country heterogeneity in health expenditure

The average per capita health spending by country, ranked in descending order, is displayed in Figure 1. The United States is the highest-spending country, with an average amount of nearly \$9,000 per capita—almost 40% higher than the second-ranked country, Switzerland. The next tier of high-spending states includes Norway, Germany, Luxembourg, and Austria, all of which allocate above \$5,000 per capita for healthcare.



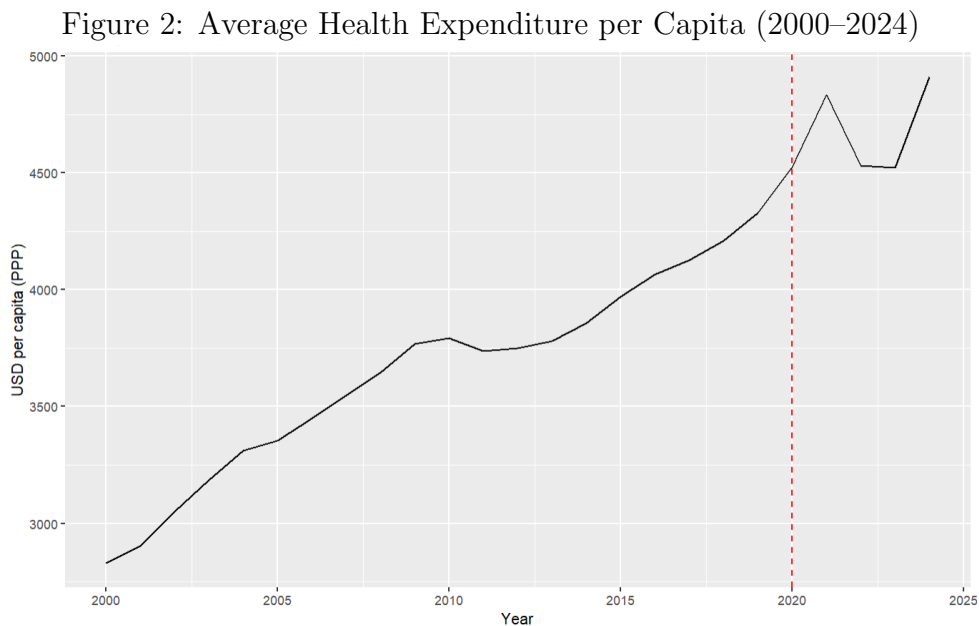
Note: Unit: USD per capita (PPP).

Source: Own compilation based on OECD Health Statistics.

Colombia, Turkey, Mexico, Costa Rica, and Latvia stand at the bottom of the distribution, contributing below \$2,000 per capita. Health spending per person varies greatly between countries, spanning a wide range from the lowest to the highest. As a result, controlling for each country’s fixed characteristics is vital in the econometric model. By using fixed effects, income elasticity is estimated based on changes within the same country over time, rather than by comparing different countries to each other. Otherwise, the coefficient could be severely biased because richer countries usually invest more financial resources in health spending, which would cause the income factor to overstate its true causal effect.

4.3 Time trends

Figure 2 presents average health spending per person for the 38 OECD countries over the period 2000–2024. There are two cycles separated by a red dashed line. Before the pandemic, from 2000 to 2019, spending on healthcare increased rapidly from approximately \$2,800 to \$4,200 per person. During 2008–2012, the growth rate



Source: Own compilation based on OECD Health Statistics.

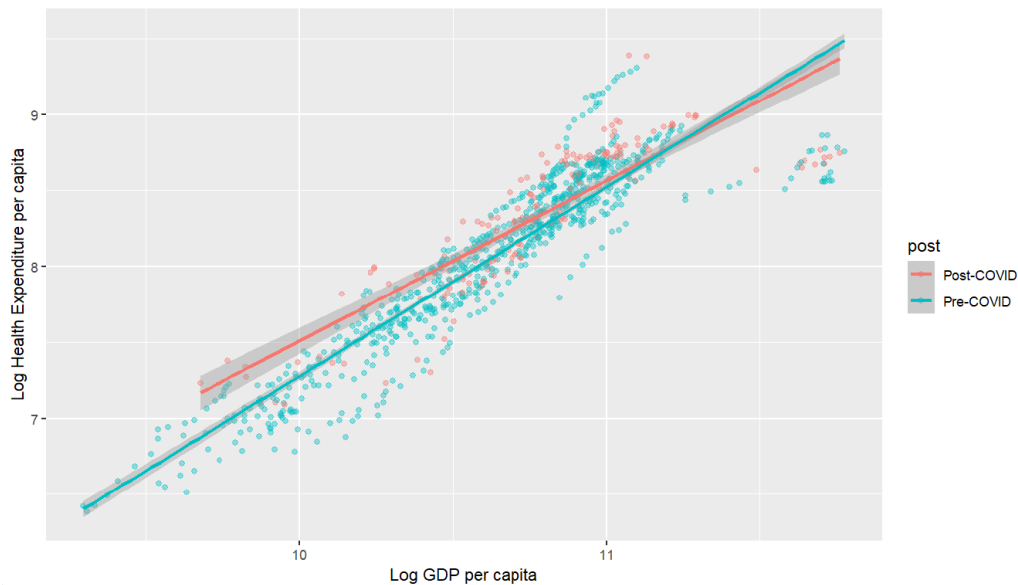
flattened as many countries reduced their budgets due to the global financial crisis.

Resources for healthcare expanded significantly after 2020, rising to almost \$4,800 per person in 2021 and marking the largest one-year increase. There is a slight fall to \$4,500 during 2022–2023, before rising again toward approximately \$5,000 by 2024. The consumption trend is not consistent with the pre-2020 trajectory; spending remains higher than what would be predicted if the 2000–2019 period had continued. This indicates the existence of a level shift after the pandemic, the relationship of which—between income and health expenditure—is the primary objective of the analysis.

4.4 The income and health expenditure relationship before and after COVID-19

Figure 3 demonstrates a scatter plot of log health expenditure per capita against log GDP per capita for 38 countries. Pre-COVID-19 data points are marked in teal, while post-pandemic data are represented by pink points, with two distinct linear regression lines for each period. In both scenarios, higher-income countries incur

Figure 3: Scatter relationship between Log Health Expenditure and Log GDP per Capita



Source: Own compilation based on OECD Health Statistics

greater expenses on healthcare, and these data points follow the respective regression lines. This suggests that a log-log relationship is an appropriate functional form and that income is a robust predictor of health spending. Additionally, the post-pandemic fitted line is clearly situated above the pre-pandemic line across all income levels. This illustrates that healthcare spending after 2020 is higher than it was prior to 2020 for any given level of GDP per capita, which is consistent with the upward shift in spending levels presented in Figure 2. Nevertheless, the slopes of the two fitted lines are essentially identical, suggesting that the spending adjustment following the COVID-19 pandemic constitutes a level effect rather than a fundamental change in the relationship between income and health expenditure. It should be noted that this visualization is based on raw data and does not yet account for within-country variation or fixed effects.

5 Econometric Model

5.1 Panel data

The main barrier to evaluating the effect of income on health spending across OECD countries is the existence of unobserved heterogeneity. These factors, such as institutional quality, the healthcare system, and citizen perception toward healthcare, are different and not easy to measure directly in each country. However, they are correlated and affect income and health expenditure simultaneously. If those factors are not considered, it would overestimate the income elasticity due to omitted variable bias. To solve this concern, through a demeaning transformation, the country fixed effects estimator uses only within-country variation by eliminating a complete time-invariant variable. Therefore, the remaining within-country variation in GDP per capita is independent of the error term, ensuring the strict exogeneity assumption is not violated. Other than factors affecting within-country variation, the model also uses two-way fixed effects by considering year fixed effects. In practice, there are unexpected common shocks such as oil price fluctuations, pandemics, and global economic turbulence that impact all OECD member states simultaneously. Year fixed effects absorb factors that modify continuously over time and affect all countries. By controlling for these common time shocks, the year fixed effects ensure that the estimated income elasticity reflects country-specific deviations from the cross-country average in each year, rather than being driven by shared macroeconomic or epidemiological trends. Therefore, the two-way model is efficient to estimate the income elasticity of health expenditure because it removes omitted variable bias by isolating the within-country and within-year correlations with GDP and health

spending.

5.2 Main model

The baseline econometric model is specified as follows:

$$\log(\text{Health}_{it}) = \alpha_i + \delta_t + \beta_1 \log(\text{GDP}_{it}) + \beta_2 [\log(\text{GDP}_{it}) \times \text{Post}_t] + \beta_3 \text{Age65}_{it} + \varepsilon_{it} \quad (1)$$

where i represents countries ($i = 1, \dots, 38$) and t denotes years ($t = 2000, \dots, 2024$). The variables are defined as follows. $\log(\text{Health}_{it})$ denotes the log of health expenditure per capita at constant prices. $\log(\text{GDP}_{it})$ denotes the log of real GDP per capita. Post_t is a binary variable equal to 1 for the period 2020 to 2024 and 0 for otherwise. Age65_{it} is the share of the population aged 65 and above. α_i represents country fixed effects, δ_t represents year fixed effects, and ε_{it} is the idiosyncratic error term.

The log-log specification is a natural choice for this model. The coefficient β_1 can be interpreted directly as an income elasticity of health expenditure before COVID pandemic: a one percent increase in GDP per capita is associated with a β_1 percent change in health expenditure. Furthermore, this specification mitigates the influence of outliers and reduces heteroskedasticity arising from cross-country differences in the level of economic development.

5.3 Interaction Term

The income elasticity of health expenditure before COVID-19 is defined as β_1 when $\text{Post}_t = 0$. The interaction term β_2 indicates the change in elasticity after 2020. As a result, the sum of the two coefficients is the income elasticity in the post-pandemic period. If β_2 is negative, it suggests that the response of health expenditure to income is weaker after COVID-19 times; if positive, it suggests a strengthening. Because year fixed effects already control for each year separately, Post_t as a standalone variable would be unessential and could introduce multicollinear problems. Additionally, the Wald test is implemented using the `linearHypothesis` function with clustered standard errors. With an aim to examine whether health expenditure is consistently responsive to income in the post-pandemic period. The null hypothesis of the test is:

$$H_0 : \beta_1 + \beta_2 = 0 \quad (2)$$

5.4 Control Variable

The model includes one control variable, which is the share of the population aged 65 and above ($Age65_{it}$). As the older population demands intensive healthcare services, population ageing is an efficient determinant of health expenditure. In addition, the aging variable is a time-consuming process determined by former fertility and mortality, making it an appropriate factor to control for when assessing health expenditure in the short run.

Other potential controls, such as unemployment rate and Government debt-to-GDP, are excluded. The unemployment rate was omitted because it performs as a mediator rather than a confounder. The fall of GDP would typically increase the unemployment rate, which partially affects health spending. According to Angrist and Pischke (2009), including the unemployment rate would absorb the causal effect and introduce "bad control" bias for the elasticity. Government debt-to-GDP was not selected because GDP is present in its denominator, generating a correlation with the regressor and potentially leading to endogeneity issues.

5.5 Standard Error

Clustering standard errors is essential in the model at the country level. Observations in each country are correlated across different years. Health spending in a specific country i is correlated with health spending in country i in year $t - 1$. Clustering at the country level allows for arbitrary within-country correlation over time. Therefore, the standard error is more robust to serial correlation and heteroskedasticity.

6 Results

6.1 Pre-COVID Baseline (2000–2019)

Table 2 reports the results of the two-way fixed effects model estimated on the pre-pandemic subsample of 38 OECD countries over the period 2000–2019 ($N = 736$). All standard errors are clustered at the country level to account for within-country serial correlation and heteroskedasticity.

The β_1 coefficient presents the income elasticity of health expenditure, which is 0.931 and significant at the 0.1% level. The estimate is below 1, suggesting that health expenditure behaves as a necessity good in OECD countries during the pre-pandemic period. A one percent increase in real GDP per capita is associated with

Table 2: Pre-COVID Period Regression Results (2000–2019)

	Estimate	Std. Error	<i>t</i> -value	<i>p</i> -value
$\log(\text{GDP}_{it})$	0.9314	0.1152	8.081	0.000***
Age65_{it}	0.0238	0.0180	1.319	0.195
Observations	736			
RMSE	0.0698			
Adjusted R^2	0.9850			
Within R^2	0.5648			

Note: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Source: Own compilation based on OECD Health Statistics.

an approximately 0.93 percent increase in health expenditure, while other factors are constant. The finding is consistent with Chakroun (2010) dealing with panel data, concluding that the income elasticity is below near one.

Share of population aged 65 and above has a positive coefficient (0.024) but presents no significant statistical difference ($p = 0.195$). While population ageing remains a structural driver of health expenditure, the country and year fixed effects absorbed its effect in this model. R-squared of the model before COVID time is 0.565, indicating that after accounting for time-invariant variables, log GDP per capita and Age65_{it} together explain about 56.5% of the within-country and cross-sectional variation of log health expenditure in the same year.

6.2 Full Panel with Interaction Term (2000–2024)

Table 3 demonstrates the estimated coefficients on the full panel data of 38 OECD countries over 2000–2024 ($N = 913$). This model includes the interaction term between log GDP per capita and the Post_t dummy variable. Therefore, the change in income elasticity of the post-pandemic period differentiates pre-pandemic and post-pandemic periods within a single model.

The income elasticity before COVID time is defined as the coefficient of log GDP per capita, which is 0.897 and highly significant ($p < 0.001$). It is a slight reduction from the result of the model during the pre-COVID time. A negative coefficient has been recognized in the interaction term ($\log(\text{GDP}_{it}) \times \text{Post}_t$), which is -0.082 , indicating a decrease in elasticity after the pandemic. Nevertheless, it is not significant at the 5% level. Therefore, the coefficient is not distinguished from zero in spite of the fall in income elasticity. Although the coefficient on Age65 is

Table 3: Main Model Regression Results (2000–2024)

	Estimate	Std. Error	<i>t</i> -value	<i>p</i> -value
$\log(\text{GDP}_{it})$	0.8975	0.1175	7.640	0.000***
$\log(\text{GDP}_{it}) \times \text{Post}_t$	−0.0820	0.0555	−1.476	0.148
Age65_{it}	0.0219	0.0147	1.485	0.146
Observations	913			
RMSE	0.0751			
Adjusted R^2	0.9822			
Within R^2	0.5907			

Note: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Source: Own compilation based on OECD Health Statistics

statistically insignificant in both circumstances, keeping it would reduce the risk of omitted variable bias. To evaluate the sensitivity of the main result to this choice, the second robustness check in section VI re-estimates the model without Age65 to verify that the income elasticity remains stable.

The test to check the post-pandemic elasticity is different from zero has been conducted by a Wald test with the null hypothesis $H_0 : \beta_1 + \beta_2 = 0$. A chi-squared statistic is 28.30 ($p < 0.001$), rejecting the null hypothesis, confirming that the responsiveness of health expenditure to income is maintained, although it is slightly weaker.

Generally, there are two findings from the test results. Firstly, health expenditure in OECD countries follows a necessity good in two circumstances before and after COVID-19, with income elasticity of 0.931 and 0.815, respectively, both below unity. The change in post-COVID elasticity of 0.815 is persistently significant and different from zero, confirmed by a linear hypothesis test. Secondly, the slight reduction in the interaction term after 2020 is not significant at a confidence level of 95%. It means that the available data do not provide adequate observations to clarify a structural break in the relationship between health expenditure and income, indicating the possibility of the limited number of post-COVID observations.

7 Robustness Analysis

7.1 Lagged GDP Per Capita

There is a possibility that GDP per capita and health expenditure are jointly determined in the same year. During the COVID time, health spending was used to pay healthcare workers overtime, purchase more equipment, and build testing centers. All these activities would increase the share of GDP in OECD countries, reflecting the effect on both variables at the same time. Therefore, the reverse causality is triggered, overestimating the elasticity.

To address this issue, instead of using the current year's income, the model is re-estimated with GDP per capita from last year. Because the prior year's income has been decided independently from this year's spending. As a result, it cannot be impacted by current health expenditure, helping to reduce the bias. This approach is consistent in practice based on backward rational expectation. Governments approve health budgets according to the prior year's economic conditions rather than on real-time figures.

The lagged model is defined as follows:

$$\log(\text{Health}_{it}) = \beta_1 \log(\text{GDP}_{i,t-1}) + \beta_2 [\log(\text{GDP}_{i,t-1}) \times \text{Post}_t] + \beta_3 \text{Age65}_{it} + \alpha_i + \delta_t + \varepsilon_{it} \quad (3)$$

38 observations are dropped as there is no information about the lagged value of the first year in each country, lowering the estimation sample from 913 to 875 observations. Table 4 presents the summary results of the main model and lagged time. The pre-COVID income elasticity resulting from lagged GDP is 0.885 ($p < 0.001$), closely compared to the estimate of 0.897 from the main model. The lagged coefficient should have declined precisely if the true effect of reverse causality existed. However, the barely any movement in elasticity implies that reverse causality does not affect the change of income elasticity, which means the rise in GDP current year is not stemmed from a jump in health expenditure. Moreover, the interaction term is -0.077 , suggesting a post-COVID elasticity of around 0.808. In both lagged and main models, post-COVID elasticity is a modest reduction. A Wald test has been implemented on the lagged model to check whether the post-COVID income elasticity is distinguished from zero. A chi-squared test confirms rejecting the null hypothesis with a statistic of 28.64 ($p < 0.001$). It indicates that when using lagged GDP to control for reverse causality, the responsiveness of health expenditure to income stands significantly.

Table 4: Main Model and Lagged GDP Robustness Check

	Main	Lagged
$\log(\text{GDP}_{it})$	0.897*** (0.117)	
$\log(\text{GDP}_{it}) \times \text{Post}_t$	-0.082 (0.056)	
Age65_{it}	0.022 (0.015)	0.022 (0.015)
$\log(\text{GDP}_{it-1})$		0.885*** (0.116)
$\log(\text{GDP}_{it-1}) \times \text{Post}_t$		-0.077 (0.053)
Num. Obs.	913	875
R^2	0.983	0.984
R^2 Adj.	0.982	0.983
R^2 Within	0.591	0.591
R^2 Within Adj.	0.589	0.590
FE: Country	X	X
FE: Year	X	X

$+p < 0.1$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$

Source: Own compilation based on OECD Health Statistics

7.2 Remove the Share of population from 65 years and over

For the second robustness check, the main model is examined without the share of the population aged 65 and above to assess the sensitivity of the income elasticity. As mentioned in Section V, the ageing variable is not significant in both circumstances. However, to prohibit the effect of omitted variable bias, keeping the ageing variable is essential because it is obviously a common cause of income levels and health expenditure. The choice to preserve the control variable has been justified by causal reasoning rather than its significance at the confidence level, illustrated by Angrist and Pischke (2009) in *Mostly Harmless Econometrics*.

Table 5: Income Elasticity With and Without Age65 Control

	$\log(\text{GDP}_{it})$ Estimate
With Age65	0.8975
Without Age65	0.9378

Source: Own compilation based on OECD Health Statistics

When the model excludes the aging variable, the income elasticity increases from 0.897 to 0.938, a shift of about 4%. Although this is a minor change, it clarifies the expected upward direction, as the older population influences both health expenditure and GDP per capita. As a consequence, removing the aging control variable would overestimate the coefficient capturing income elasticity. Therefore, it is an insurance factor against omitted variable bias.

7.3 Excluding 2020 and Redefining the Post-COVID Period

In 2020, GDP in OECD countries shrank significantly due to lockdowns, while health expenditure grew enormously in response to rising demand in intensive treatment healthcare. Considering 2020, an immediate response could be to see the true effect of long-term change in health expenditure. Therefore, to deal with this concern, removing 2020 from the sample and reformulating the post-COVID time in the period 2021–2024 has been conducted.

The coefficient of $\log_GDP$ demonstrating the income elasticity before COVID time is 0.905, significantly and close to the main estimate of 0.897. The interaction term is -0.089 , indicating the post-COVID elasticity is 0.816, almost similar to the post-COVID elasticity of the main model, 0.815. A Wald test verifies that post-COVID in a redefined period is different from zero.

Table 6: Robustness Check: Excluding 2020

	Estimate	Std. Error	<i>t</i> -value	<i>p</i> -value
$\log(\text{GDP}_{it})$	0.9047	0.1177	7.688	0.000***
$\log(\text{GDP}_{it}) \times \text{Post}_t$	-0.0890	0.0631	-1.411	0.167
Age65_{it}	0.0218	0.0147	1.487	0.146
Observations	875			
R^2 Adj.	0.9822			
R^2 Within	0.5905			

Note: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Source: Own compilation based on OECD Health Statistics..

8 Conclusion

The paper implements an experiment to test whether the COVID-19 pandemic has impacted the income elasticity of health expenditure in OECD countries. With the approach of a two-way fixed effects model on the panel data of 38 countries over 2000–2024, the analysis inspects the relationship between GDP per capita and health spending after the pandemic while controlling for the aging variable.

There are three main findings from the results. First, health expenditure in OECD countries behaves as a necessity good, with income elasticities below unity. The income elasticity pre-COVID time is estimated at around 0.9, suggesting that a one percent increase in GDP per capita is associated with less than a proportional increase in health spending. Second, the Wald test concludes that the post-COVID income elasticity is continually distinguished from zero, maintaining the existence of the relationship between income and health spending after the pandemic shock. Third, the minor decline in income elasticity after the pandemic is insignificant at the confidence level. It means that there is not enough information to confirm that the pandemic alters the connection between income and health expenditure.

These findings were assessed through various robustness checks. Replacing GDP per capita with one prior year lagged to address the reverse causality issue. The coefficients in the main and lagged models are basically the same, suggesting that reverse causality does not affect the change in income elasticity. Excluding the ageing control variable makes the elasticity upward, highlighting its responsibility to absorb omitted variable bias in spite of being statistically insignificant. Redefining the post-pandemic period by removing 2020 shows barely any shift in income elasticity. It suggests that the coefficient is not driven by the following crisis year.

On the other hand, there are some limitations in conducting the study. The main limitation is the shortage of post-COVID year observations with a 5-year panel data from 2020 to 2024. As a result, the model cannot have adequate statistical power to identify a structural break. Moreover, while two-way fixed effects regulate time-invariant country characteristics, it does not completely explain the endogeneity of GDP. Finally, the panel is unbalanced because of data unavailability in some countries, which may imprecisely estimate the post-COVID elasticity.

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